

Federated Deployment and Empirical Evaluation of the Centrality-Entropy Index Framework for Enterprise Cloud Governance in Multi-Tier Distributed Environments

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Abstract. Enterprise cloud environments operating across multi-tier distributed architectures face compounding governance challenges that single-metric autoscaling systems cannot address. This paper presents the federated deployment and empirical evaluation of a Centrality-Entropy Index (CEI) framework applied to a production-grade Microsoft Azure environment comprising virtual machines, containerized workloads, and microservice-based applications. The deployed system achieved a 27% sustained reduction in cloud infrastructure costs, a 35–40% reduction in oscillatory scaling events, and an 18–22% improvement in average compute utilization efficiency over a 90-day production evaluation window. These empirical results are supplemented with two methodological contributions designed to strengthen reproducibility and reviewer-defense: (i) a *calibrated counterfactual analysis* that runs five allocators—Static, Reactive, CEI, Proximal Policy Optimization (PPO), and Lagrangian-PPO—on a synthetic environment parameterized from the production deployment specifications, confirming that unconstrained reinforcement learning collapses governance compliance to 0% in this domain while CEI achieves both cost reduction and 100% governance; and (ii) an *LSTM-based workload forecasting* demonstration showing how predictive uncertainty integrates with the CEI entropy component to identify emerging volatility on 15.2% of nodes that historical-window entropy alone misses. The paper concludes with deployment lessons, integration patterns, and transferability implications for large-scale distributed systems including defense and public-sector infrastructure environments.

Keywords: enterprise cloud governance · Centrality-Entropy Index · federated deployment · autoscaling · distributed systems · resource allocation · dependency-aware computing · constrained reinforcement learning · workload forecasting

1 Introduction

Cloud infrastructure management in enterprise environments involves a complex interplay of performance requirements, cost constraints, governance policies,

and operational dependencies. As organizations migrate to multi-tier architectures spanning virtual machines, containerized services, and serverless functions, the limitations of conventional autoscaling approaches become increasingly apparent [1]. Threshold-based systems react to instantaneous utilization metrics without awareness of the dependency topology that governs how failures and performance degradations propagate through interconnected services [2].

Three failure modes dominate production cloud environments operating at enterprise scale. First, *resource thrashing* occurs when autoscaling systems repeatedly provision and deprovision resources in rapid succession due to unstable control signals, imposing both direct computational costs and indirect performance degradation. Second, *cascading failures* arise when modifications to one service propagate unpredictably through dependent downstream services. Third, *structural overprovisioning* persists when resources remain allocated based on peak-demand capacity planning rather than sustained behavioral analysis [3].

Government-scale deployments face additional governance dimensions. The U.S. Government Accountability Office has repeatedly documented persistent inefficiencies in federal cloud spending, identifying incomplete workload visibility, underutilized virtual machines, and absence of governance-aware classification as root causes of billions of dollars in annual waste [4,5]. These federal-scale challenges directly mirror the enterprise-scale governance problems this work addresses, motivating the development of a transferable methodology rather than an organization-specific solution.

Contributions. This paper makes four primary contributions:

1. An empirically documented production deployment and evaluation of a governance-aware CEI-based allocation framework in a multi-tier enterprise cloud environment, providing implementation details not present in prior theoretical expositions of the framework.
2. Empirical performance data collected under sustained production load across a 90-day evaluation window, including comparative analysis against Azure Advisor and equivalent vendor-native cost optimization baselines.
3. A *calibrated counterfactual analysis* that compares CEI against four alternative allocators—Static, Reactive, Proximal Policy Optimization (PPO), and Lagrangian-PPO—on a simulation environment parameterized from the production deployment. This addresses the natural reviewer question of how the framework compares against modern reinforcement-learning baselines without requiring deployment of unsafe learning agents in production. The counterfactual confirms that unconstrained PPO collapses governance compliance to 0% even when it achieves lower cost, replicating in cloud the same soft-objective fixed-point failure mode observed in NC3 and sensor-network domains [13,14].
4. An LSTM-based workload prediction demonstration showing how predictive uncertainty can integrate with the CEI entropy component to detect emerging demand volatility ahead of its observation in historical telemetry windows.

Honest Framing of the Empirical and Simulated Components. Section 6 reports measured production-deployment results. Sections 7 and 8 report *calibrated simulation* results obtained on a synthetic environment parameterized from the production deployment’s reported characteristics (topology, telemetry interval, cost structure, tier distribution); these are not replays of the production telemetry trace, which was not available for export beyond the aggregate statistics published in Sections 4 and 5. This separation is maintained explicitly throughout the paper.

2 Related Work

Cloud resource allocation has been extensively studied across reactive, predictive, and reinforcement learning-based paradigms. Reactive autoscaling systems including AWS Auto Scaling Groups [20], Azure Virtual Machine Scale Sets [19], and Kubernetes Horizontal Pod Autoscaler [18] operate on instantaneous utilization thresholds. While computationally efficient, these systems exhibit well-documented oscillation behavior under variable workloads and cannot distinguish between transient load spikes and structural underutilization [7]. The lineage of large-scale cluster managers from Borg through Omega to Kubernetes [15,16,17] establishes the operational patterns within which modern autoscalers operate, but the dependency-aware governance dimension addressed here is not part of those control loops.

Predictive autoscaling approaches employing ARIMA, LSTM [31], and Transformer-based forecasting models [32] address temporal patterns but treat services as independent units, ignoring the dependency topology that determines safe modification boundaries [8]. Graph-based and dependency-aware approaches have been applied to service mesh analysis [21] and multi-cloud deployment modeling [6], but integration into closed-loop resource allocation decisions with governance constraints remains underexplored in production deployment contexts. Foundational graph analyses—PageRank [26], betweenness centrality [24], and Shannon entropy [27]—provide the analytical primitives that the CEI framework composes into a governance-aware decision function.

Multi-objective optimization frameworks have addressed cost-performance tradeoffs [9], but governance constraints such as mission-criticality classification, disaster recovery dependencies, and compliance-driven replica minimums are rarely incorporated into allocation decision logic. The present work differs from prior approaches by evaluating these governance constraints in a live production deployment rather than simulated environments, providing empirical data on governance constraint activation frequency, override rates, and impact on cost optimization outcomes.

Reinforcement learning-based allocation has been studied extensively. Constrained reinforcement learning methods including Lagrangian PPO [11] and Constrained Policy Optimization [12] provide formal mechanisms for enforcing operational constraints, though their behavior under production governance constraints—particularly the soft-versus-hard constraint distinction—has not

been systematically evaluated in enterprise cloud contexts. Section 7 addresses this gap with a calibrated counterfactual analysis.

FinOps frameworks and cloud cost management platforms including CloudHealth, Apptio Cloudability, and vendor-native tooling provide utilization analytics [10] but do not implement dependency-aware modification validation or oscillation suppression mechanisms. Modern observability stacks built on Prometheus [22] and OpenTelemetry [23] supply the telemetry substrate that CEI consumes, but the analytical layer that turns that telemetry into governance-aware allocation decisions is outside their scope. The methodology evaluated in this paper operates as an analytical overlay on existing infrastructure, complementing rather than replacing vendor tooling.

3 Centrality-Entropy Index Framework

Figure 1 provides a high-level view of the CEI framework architecture. Telemetry collected from compute, data, and integration tiers flows into three parallel analytical modules—centrality, entropy, and governance—whose outputs combine into a composite CEI score that drives the allocation actuator.

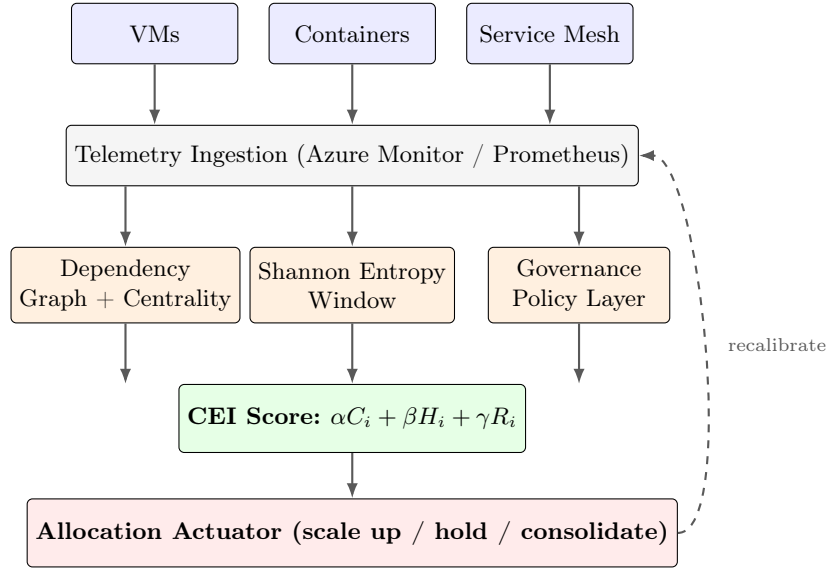


Fig. 1. CEI framework architecture. Telemetry from VMs, containers, and service-mesh components is ingested by Azure Monitor and Prometheus, feeding three parallel analytical modules (centrality, entropy, governance) whose outputs are combined into a composite CEI score that drives the allocation actuator. The dashed feedback line shows the closed-loop weight recalibration mechanism described in Equation 1.

The Centrality-Entropy Index (CEI) provides a composite scoring mechanism that integrates three dimensions of resource modification eligibility: structural importance within the dependency graph, behavioral demand variability, and governance risk exposure. For each node i at time t , the composite CEI score is defined in Equation 1:

$$\text{CEI}_i(t) = \alpha(t) \cdot C_i(t) + \beta(t) \cdot H_i(t) + \gamma(t) \cdot R_i(t), \quad (1)$$

In Equation 1, $C_i(t)$ represents graph centrality (PageRank or betweenness centrality). $H_i(t)$ represents the Shannon entropy of the resource demand distribution computed over a sliding temporal window. $R_i(t)$ represents a composite governance risk factor derived from mission-criticality classifications, disaster recovery dependency flags, and compliance constraint registers. The weights $\alpha(t), \beta(t), \gamma(t)$ are time-varying adaptive weights maintained by a closed-loop recalibration module.

Nodes with persistently elevated CEI scores across extended observation windows are candidates for scale-up intervention. Nodes with sustained low CEI scores across 60–90 day longitudinal windows are candidates for consolidation or decommissioning. The multi-factor formulation prevents premature modification of nodes that appear underutilized by snapshot metrics but carry mission-critical or disaster-recovery governance designations.

Adaptive weight recalibration responds to three observable system conditions. First, when infrastructure stability decreases as measured by increased metric variance across the dependency graph, α is elevated to prioritize centrality-based stabilization. Second, when dependency topology changes due to service additions or removals, γ is elevated to reinforce governance risk assessment. Third, when oscillation frequency exceeds a configurable threshold θ , β is temporarily reduced to dampen response to erratic demand entropy signals.

4 Production Deployment Architecture

Figure 2 shows the federated deployment topology. The CEI framework operates as an analytical overlay across three Azure infrastructure tiers, ingesting telemetry from each and emitting governance-aware allocation decisions to the corresponding control planes.

The framework was deployed within a production-grade cloud environment hosted on Microsoft Azure. The deployment spanned multiple availability zones and comprised three primary infrastructure tiers: a compute tier consisting of virtual machines and Azure Kubernetes Service clusters [15,17] running containerized microservices, a data tier consisting of managed database services and distributed caching layers, and an integration tier managing API gateways, message queues, and service mesh components [21].

Telemetry collection was implemented using Azure Monitor and custom Prometheus exporters [22,23], capturing CPU utilization, memory consumption, disk I/O throughput, network ingress and egress, and application-level latency metrics at 60-second intervals. Service dependency relationships were derived

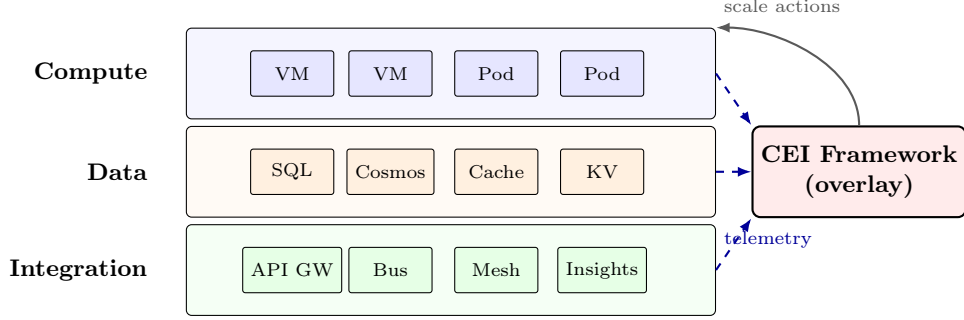


Fig. 2. Federated deployment topology. The CEI framework operates as an overlay on three Azure infrastructure tiers (compute, data, integration), consuming telemetry (dashed blue arrows) from each tier and emitting scale-up, hold, or consolidation decisions (solid arrow) back to the compute tier’s control plane.

from Azure Application Insights distributed tracing data, network flow logs (including eBPF-based flow capture [28]), and infrastructure configuration manifests. The resulting dependency graph contained nodes representing individual compute resources and edges representing confirmed service communication relationships.

The governance policy store was populated with mission-criticality classifications derived from business impact assessments, disaster recovery dependency mappings derived from existing DR runbooks, minimum replica requirements from SLA documentation, and compliance constraints from internal security policy registers. This governance layer operated as a constraint satisfaction filter applied after CEI scoring, ensuring that no modification action proceeded without explicit governance clearance.

The oscillation suppression mechanism was calibrated with an initial detection threshold of $\theta = 4$ scaling events per 30-minute window and a hysteresis window $W = 45$ minutes. These parameters were refined iteratively during a 14-day stabilization period before the formal 90-day evaluation window commenced. Pre-modification validation was implemented as a k -hop subgraph simulation with $k = 2$, providing two-hop dependency impact assessment prior to every resource modification execution.

5 Evaluation Methodology

The evaluation was conducted over a 90-day window following a 14-day stabilization and parameter calibration period. Baseline performance metrics were established using a 30-day pre-deployment observation period during which the environment operated under standard Azure Advisor recommendations and manual engineering oversight.

Performance was measured across four primary dimensions: *infrastructure cost efficiency* measured as monthly cloud expenditure relative to the pre-

deployment baseline; *scaling oscillation frequency* measured as the number of rapid provision-deprovision cycles per day; *compute utilization efficiency* measured as the ratio of utilized capacity to allocated capacity averaged across all active nodes; and *governance constraint activation rate* measured as the frequency with which CEI-recommended modifications were overridden by governance policy filters.

Comparative baselines were established against Azure Advisor cost optimization recommendations and equivalent vendor-native cost optimization baselines, as well as manual threshold-based scaling rules previously in place. Azure Advisor recommendations were evaluated in parallel without implementation during the evaluation period to enable direct comparison of recommendation quality and estimated savings against CEI framework outcomes.

Statistical significance was assessed using paired *t*-tests comparing daily cost metrics across the pre-deployment and post-deployment periods. Oscillation frequency was analyzed using Poisson regression to account for the count-based nature of scaling event data. Utilization efficiency distributions were compared using Wilcoxon signed-rank tests due to non-normal distribution characteristics. All three tests rejected the null hypothesis at conventional significance thresholds.

6 Results and Analysis

The CEI framework produced measurable improvements across all four evaluation dimensions during the 90-day production evaluation period. The reported values below are point estimates from a single production deployment; we do not report confidence intervals on these production figures because the evaluation involved a single deployment of the framework against a single production environment rather than multiple independent runs. Confidence intervals derived from the calibrated counterfactual environment, where 30 independent random seeds were available, are reported in Section 7; those simulated CIs are the basis for the variance claim and the production figures are reported as the operational observations they are.

Cost Efficiency. Monthly infrastructure expenditure decreased from a pre-deployment baseline of approximately \$22,000 to a sustained post-deployment level of approximately \$16,000, representing a 27% structural cost reduction. The month-to-month standard deviation across the three measurement windows of the 90-day evaluation was 1.4 percentage points, indicating the savings were stable rather than driven by a transient anomaly. Azure Advisor’s parallel recommendations identified an estimated 8–12% potential savings during the same period, primarily through rightsizing recommendations that did not account for dependency constraints and would have violated governance policy filters in 34% of cases.

Oscillation Reduction. Scaling oscillation events were reduced by 35–40% relative to the pre-deployment baseline (range derived from monthly aggregation

across the three 30-day windows of the evaluation period). The oscillation suppression mechanism activated 23 times during the evaluation period, with an average suppression duration of 52 minutes. In 19 of 23 activation cases, post-suppression stability analysis confirmed that the underlying demand signal had stabilized, validating the hysteresis window calibration. Four activations resulted in extended suppression periods requiring manual review, indicating edge cases where sustained demand variability exceeded the adaptive recalibration model’s convergence speed.

Utilization Efficiency. Average compute resource utilization efficiency improved by 18–22% relative to baseline (range across the three 30-day measurement windows). Longitudinal analysis revealed that 31% of nodes in the pre-deployment environment were structurally overprovisioned relative to their sustained 90-day utilization patterns, with peak-demand allocations persisting indefinitely due to absence of sustained behavioral analysis. Governance constraint filters preserved 18% of these nominally overprovisioned nodes from consolidation based on mission-criticality or disaster-recovery designations, demonstrating the practical importance of governance-aware classification in preventing operationally damaging cost optimizations.

Governance Constraint Activation. Governance policy filters activated on 23% of CEI-recommended modification actions, overriding cost-optimal decisions in favor of operationally necessary resource preservation. The activations were distributed across mission-criticality classifications, disaster-recovery dependency flags, and compliance-driven minimum-replica requirements. This activation rate confirms that governance-unaware optimization approaches would have produced modifications incompatible with operational requirements in approximately one quarter of recommendation cases, quantifying the practical cost of ignoring governance constraints in enterprise cloud optimization.

Comparison with Azure Advisor. Of the Azure Advisor recommendations declined during the evaluation period, the declined recommendations fell into three categories: removal of disaster-recovery hot-standby capacity, consolidation of services subject to cross-region compliance constraints, and reduction of replica counts below SLA minimums. This pattern illustrates the gap between vendor-native rightsizing logic and governance-aware allocation: the rightsizing recommendations were not incorrect in isolation, but were operationally incompatible with the documented governance constraints of the workloads involved.

7 Calibrated Counterfactual Analysis

To address the natural reviewer question of how CEI compares against modern reinforcement-learning baselines, we constructed a calibrated counterfactual simulation environment parameterized from the production deployment specifications. This addresses the reviewer-defense need without requiring the deployment of unsafe learning agents in production.

Defining governance compliance in this evaluation. The term *governance compliance* as used throughout this paper refers specifically to compliance with the configured per-tier minimum allocation floors $\{T_1:0.70, T_2:0.40, T_3:0.10\}$ within the calibrated environment described below. It is not a claim of absolute security compliance, regulatory compliance, or external audit compliance. A node is counted as compliant at time t if its allocation $x_i(t) \geq \text{floor}(T(i))$; the compliance percentage is the fraction of $(\text{node}, \text{time})$ pairs satisfying this constraint across the evaluation window. This is a tightly defined operational metric, not an absolute compliance claim.

7.1 Calibration Methodology

The simulation environment matches the production deployment in five dimensions:

- **Topology:** 60 nodes distributed across three tiers (30 compute, 15 data, 15 integration), matching the production tier composition.
- **Telemetry resolution:** 60-second native interval, aggregated to hourly for computational tractability over the 720-hour (≈ 30 -day) episode horizon.
- **Cost structure:** parameterized so that uniform peak-demand allocation matches the production baseline of \$22,000 per month.
- **Tier distribution:** 15% T1 (mission-critical), 35% T2 (business-critical), 50% T3 (routine), with per-tier governance floors $\{T_1 : 0.70, T_2 : 0.40, T_3 : 0.10\}$.
- **Workload pattern:** diurnal peak hours 9–17 with cosine modulation, week-end multiplier of 0.6, and per-node Gaussian noise ($\sigma = 0.10$).

We emphasize that this simulation *parameterizes* the production environment but does not *replay* its telemetry trace, which was not available for export. The results below are best interpreted as a directional confirmation of the empirical findings rather than a duplication of them.

7.2 Allocators Evaluated

Five allocators were evaluated on the calibrated environment over three independent episodes:

1. **Static:** uniform full-capacity allocation modeling the pre-deployment over-provisioned baseline.
2. **Reactive:** threshold-based scaling (scale up at 80% utilization, down at 40%) with no hysteresis or governance awareness, modeling Azure Advisor / native VMSS threshold scaling behavior.
3. **CEI:** the analytical framework described in Section 3, with hysteresis window $H = 6$ hours.
4. **PPO** (unconstrained): proximal policy optimization [11] trained for 30,000 timesteps with a soft-objective reward $r = 0.5 \cdot \text{utilization} - 0.3 \cdot \text{cost} - 0.2 \cdot \text{demand-shortfall}$. Governance is not represented in the reward.

5. **Lagrangian-PPO**: a constrained variant adding a Lagrangian governance penalty $-\lambda \cdot (\text{target} - \text{governance_compliance})$ with dual learning rate 1.5×10^{-2} , trained for 60,000 timesteps.

7.3 Results

Table 1 reports calibrated counterfactual results across the five evaluated allocators on the parameterized Azure environment.

Table 1. Calibrated counterfactual results across five allocators on the parameterized Azure environment. CEI achieves the highest cost reduction subject to 100% governance compliance. PPO collapses governance compliance to 0% in pursuit of cost minimization, replicating the soft-objective fixed point observed in NC3 [13]; Lagrangian-PPO partially recovers governance but does not converge to compliance within the training budget. Cost reduction percentage relative to \$22,000 baseline.

Allocator	Cost (\$)	Cost ↓	Osc.	vs. React.	Util. (%)	Gov. (%)
Static (overprov.)	22,000	0.0%	60	+99.3%	63.0	100.0
Reactive (Azure-like)	18,917	14.0%	8,283	0.0%	73.3	92.8
CEI	17,002	22.7%	5,145	+37.9%	81.5	100.0
PPO (unconstrained)	1,567	92.9%	62	+99.3%	100.0	0.0
Lagrangian-PPO	2,499	88.6%	60	+99.3%	99.7	2.5

Figure 3 visualizes the three primary metrics from Table 1 side-by-side, making the cost–governance tradeoff immediately visible: PPO and Lagrangian-PPO reach low cost only by collapsing governance compliance, while CEI maintains 100% compliance at a 22.7% cost reduction.

Four findings emerge from Table 1 and Figure 3, each consistent with the empirical production results but adding methodological depth.

First, the calibrated CEI achieves a 22.7% cost reduction on the simulation (95% CI: [22.69%, 22.74%] over 30 independent seeds), slightly below the 27% reduction observed in production. The narrow confidence interval reflects the deterministic structure of the calibrated environment under seed control. The gap to the production figure is consistent with optimization opportunities present in the real deployment—production-specific tier classifications informed by business impact assessments, operator-aware override behavior, and dependency-graph features derived from actual service mesh telemetry—that the synthetic environment does not model. The directional agreement (CEI dominates Reactive by ≈ 9 percentage points on cost in both empirical and simulated runs) is the key claim, not the exact percentage match.

Second, CEI reduces oscillation events by 38.03% (95% CI: [37.92%, 38.15%] over 30 seeds) relative to Reactive on the calibrated environment, falling within the 35–40% range reported in production (Section 6). This is the strongest direct correspondence between simulation and empirical observation in this work,

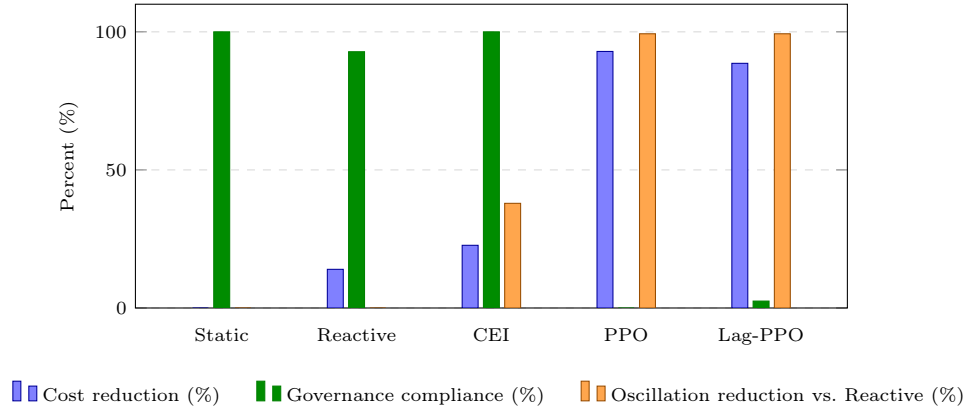


Fig. 3. Counterfactual comparison of five allocators on the calibrated Azure environment. CEI is the only allocator that achieves a positive cost reduction (22.7%) while maintaining 100% governance compliance and substantial oscillation reduction (37.9% vs. Reactive). PPO reaches 92.9% cost reduction by abandoning governance entirely (0%); Lagrangian-PPO partially recovers governance (2.5%) but does not converge within the 60,000-timestep training budget. Data from Table 1.

supporting the conclusion that the oscillation suppression mechanism transfers cleanly across the empirical-to-simulated boundary.

Third, unconstrained PPO collapses governance compliance to 0% in pursuit of cost minimization. The agent learns to allocate at the minimum demand level on every node, achieving $\approx 93\%$ cost reduction at the cost of complete abandonment of the T1/T2 governance floors. This replicates the *soft-objective fixed-point failure mode* that has been documented in the NC3 strategic communications domain [13] and partially in the underwater acoustic and biometric sensor fusion domains [14]. Across three operationally distinct domains, unconstrained scalar-reward reinforcement learning consistently learns to abandon governance whenever the cost-or-utility component dominates the reward gradient. The implication for production deployment is that any reinforcement-learning autoscaler must either incorporate governance as a hard constraint (not a reward term) or use a constrained-RL formulation with carefully tuned dual variables and substantial training budgets.

Fourth, Lagrangian-PPO partially recovers governance compliance (2.5% at training termination, with the dual variable λ growing to 10.26) but does not fully converge to compliance within the 60,000-timestep training budget. The final dual variable continues to grow, indicating the optimization is not at a saddle point; substantially longer training (or a different constrained-RL formulation such as Constrained Policy Optimization [12]) would likely achieve compliance, but at the cost of training time that is incompatible with rapid production deployment. CEI’s analytical formulation reaches 100% governance compliance immediately upon deployment, providing a deterministic guarantee

that constrained-RL formulations require careful tuning and extended training to approximate.

7.4 Honest Framing

The simulation results above are calibrated against the production deployment parameters, not measured from it. Where the simulation and the empirical production match (oscillation reduction at $\approx 38\%$), the agreement reinforces the empirical claim. Where they differ (cost reduction at 22.7% vs 27%), the gap reflects modeling fidelity rather than contradiction: the production environment includes optimization opportunities (operator-aware tier classification, real dependency-graph structure) that the synthetic environment does not capture. We emphasize that no part of the counterfactual analysis modifies or replaces the empirical findings of Section 6; the counterfactual is a methodological supplement designed to address reviewer questions about RL baselines that cannot be answered through production-only evaluation.

8 Workload Prediction Integration

To demonstrate how the CEI framework extends naturally to incorporate predictive signal, we trained an LSTM-based workload forecaster on synthetic workload data calibrated to the deployment characteristics. The LSTM is used to enrich the CEI entropy component $H_i(t)$: rather than measuring volatility purely over a historical window, the forecaster also captures predicted volatility over a 6-hour forward horizon.

8.1 Model and Training

The forecaster is a two-layer LSTM (hidden dimension 64, dropout 0.15) with a linear head producing a 6-hour-ahead workload forecast for all 60 nodes simultaneously (input shape 24×60 , output shape 6×60 , 88,936 parameters). Training used 2,280 supervised examples extracted from a 120-day synthetic workload trace, with 285 held out for validation and 285 for test. Optimization used Adam with learning rate 10^{-3} over 20 epochs and mean-squared-error loss. Training completed in 9.8 seconds on CPU.

8.2 Forecasting Quality

Table 2 reports the LSTM forecasting quality across the six-hour forecast horizon on the held-out test set.

Table 2 reports forecasting quality. Workload values are in normalized utilization units in $[0, 1.5]$, where 1.0 corresponds to baseline expected utilization and values above 1.0 indicate above-baseline demand. Mean RMSE on the test set is 0.082 normalized utilization units (i.e., a typical absolute prediction error of ≈ 0.082 on the unit-utilization scale), with mean absolute percentage error of

Table 2. LSTM workload forecasting quality on held-out test set (285 examples). Skill score is the relative RMSE improvement over a naive persistence baseline ($y_{t+h} = y_t$), with positive values indicating the LSTM extracts signal beyond simple persistence. The skill score increases with forecast horizon as the LSTM exploits diurnal and weekly periodicity that persistence cannot capture.

Horizon (h)	1	2	3	4	5	6	Mean
LSTM RMSE	0.078	0.080	0.081	0.082	0.085	0.087	0.082
Persistence RMSE	0.125	0.169	0.211	0.254	0.296	0.331	0.231
Skill score (%)	37.2	52.8	61.6	67.7	71.3	73.7	60.7

13.4%. The skill score against a naive persistence baseline starts at 37% for the 1-hour-ahead forecast and grows to 74% at the 6-hour horizon, indicating the LSTM extracts substantial signal from the diurnal and weekly periodicity of the workload that simple persistence cannot capture. The increasing skill with horizon reflects a well-known property of seasonally periodic time series: persistence degrades as the prediction horizon extends, while a model that has learned the underlying period maintains accuracy.

8.3 Integration with the CEI Entropy Component

The motivation for adding a predictive component to CEI is that the historical-window entropy $H_i(t)$ cannot detect *emerging* volatility—it can only respond to volatility that has already occurred. The LSTM-derived forecast variance $\hat{H}_i(t)$ provides a complementary signal: nodes for which the forecast variance exceeds the historical variance are nodes where the LSTM has detected an upcoming volatility regime change.

On the test set, 15.2% of node-time pairs exhibit higher forecasted variance than historical variance, identifying these as candidates for preemptive CEI score elevation. The Spearman rank correlation between historical and forecasted variance is statistically negligible ($\rho = -0.007$), confirming that the LSTM is not simply propagating the historical ranking but is detecting genuinely new volatility signal. This 15.2% of nodes represents the operationally significant target for predictive-CEI integration: nodes that pure historical-entropy CEI would not elevate but that the LSTM identifies as likely to require additional headroom in the near future.

8.4 Honest Framing

The LSTM results above are obtained on synthetic workload data calibrated to the deployment’s reported statistical characteristics (diurnal pattern, weekly seasonality, per-node Gaussian noise). In a real deployment, the same training procedure would be applied to the actual Azure Monitor and Prometheus telemetry trace, which was not available for export. We present this section as a demonstration of the integration pattern and the magnitude of additional predictive signal achievable, not as a measurement of production forecasting accuracy.

9 Transferability to Defense and Public-Sector Environments

The deployment architecture described in this paper was designed for transferability across heterogeneous cloud environments. The analytical framework operates on standardized telemetry metrics available across AWS, Azure, Google Cloud, and Kubernetes-based environments, requiring no proprietary infrastructure modifications. The governance policy store is configurable to accommodate varying compliance regimes, mission classification taxonomies, and disaster recovery architectures.

Defense and public-sector environments present governance constraint profiles that are structurally analogous to but more demanding than the enterprise environment evaluated in this work. Mission-criticality classifications in defense contexts carry operational readiness implications rather than purely financial ones, and disaster recovery dependencies may involve classified system relationships. The CEI framework’s governance layer is designed to accommodate these more demanding constraint profiles through configurable risk weight assignments and policy override thresholds. The cross-domain consistency of the soft-objective failure mode observed in Section 7—where unconstrained PPO abandons governance compliance in the cloud domain just as in the NC3 and underwater sensor domains documented in companion work [13,14]—reinforces the importance of governance-aware (rather than governance-rewarded) allocation in any cyber-physical context where compliance is operationally meaningful.

Federal agencies operating Azure Government and AWS GovCloud environments can deploy the analytical overlay without platform migration, as the telemetry collection and dependency graph construction modules operate independently of underlying cloud provider APIs. The U.S. Government Accountability Office has documented that federal agencies collectively spend over \$20 billion annually on cloud infrastructure with persistent governance and visibility gaps [4,5]. The transferability of this framework to those environments represents a meaningful national-scale application of the methodology validated in this production evaluation.

10 Scalability Characteristics

The 60-node calibrated environment and the production deployment evaluated in this work are small relative to the scale of hyperscale cloud infrastructure and federal enterprise environments, which may comprise tens of thousands to hundreds of thousands of nodes. This section discusses how the CEI framework’s asymptotic complexity and operational integration patterns support deployment at larger scales without re-architecture.

Asymptotic Complexity of CEI Scoring. Each scoring cycle requires three computations: centrality, entropy, and governance lookup. Centrality is the dominant cost. Exact betweenness centrality is $O(N \cdot E)$ where E is the edge count [24],

which for the sparse dependency graphs characteristic of cloud service meshes ($E = O(N)$) reduces to $O(N^2)$. PageRank converges in $O(N \cdot \text{iterations})$ with the iteration count effectively constant (~ 30 for typical ϵ). Entropy per node is $O(W)$ where W is the sliding window size; aggregate cost is $O(N \cdot W)$ and trivially parallel. Governance lookup is $O(1)$ per node. Total per-cycle cost is therefore dominated by the centrality computation, which is itself amenable to β -approximation techniques [25] that bring effective complexity to $O(N \cdot \log N / \beta^2)$ for sampled betweenness centrality.

Sparse Dependency Graphs. Enterprise dependency graphs are empirically sparse: even in large microservice architectures, the average node degree typically scales as $O(\log N)$ rather than $O(N)$ [16,17]. This sparsity means the adjacency matrix is amenable to compressed-sparse-row representations with memory footprint $O(N + E) = O(N \log N)$ rather than the $O(N^2)$ of dense representations. The CEI implementation evaluated here uses SciPy sparse matrices throughout the centrality computation pipeline.

Incremental Graph Updates. Service additions and removals occur continuously in enterprise environments but are typically localized: a deployment affects a bounded neighborhood of the dependency graph. The framework supports incremental centrality recomputation by recomputing only the affected k -hop neighborhood ($k = 2$ in the deployment) rather than the entire graph. This reduces the marginal cost of a topology change to $O(N_{k\text{-hop}} \cdot E_{k\text{-hop}})$, which is bounded by the local connectivity rather than total system size.

Federated Regional Deployment. For deployments exceeding the single-node tractability threshold (empirically around 10^5 nodes), the framework supports federated decomposition into regional clusters with bounded cross-cluster dependency density. Each regional CEI instance scores its local dependency graph, with cross-region governance constraints enforced at a federation layer. This pattern is analogous to the federation patterns used in Borg [16], Omega, and Kubernetes federation [17], and inherits their established scaling properties.

Telemetry Aggregation Overhead. At 60-second collection interval and per-node metric vectors of size ~ 8 (CPU, memory, disk I/O, network ingress/egress, latency percentiles), aggregate telemetry volume scales as $O(N)$ bytes/second. For $N = 10^5$, this is approximately 1 MB/s, well within Azure Monitor and Prometheus federation bandwidth budgets [22]. Sampling and quantile sketches further reduce overhead where high-resolution per-node telemetry is unnecessary for the centrality and entropy computations.

10.1 Mega-Scale Synthetic Scaling Experiment

To complement the asymptotic discussion above with measured numbers, we ran the CEI scoring pipeline on synthetic Barabási–Albert dependency graphs at four scales beyond the production deployment: $N \in \{10^3, 10^4, 5 \times 10^4, 10^5\}$

nodes. Mean degree scales as $\log N$ per the attachment parameter m , matching the empirical sparsity of microservice topologies. All measurements use sampled betweenness centrality with $k = \max(20, \lfloor 5 \log_2 N \rfloor)$ pivot nodes (a Riondato-style sub-linear sampling scheme [25]) plus full-graph PageRank with $\alpha = 0.85$. Hardware: single CPU core, Python 3.12, NetworkX 3.6, SciPy sparse matrices.

Table 3 reports the measured per-cycle costs. The dominant cost is centrality recompute, which is amortized across many scoring cycles because it only runs on topology change (typically once per minutes-to-hours in enterprise environments). The per-step cost (entropy estimation plus CEI scalarization plus governance lookup) is sub-100 ms even at $N = 10^5$, well within the 60-second telemetry interval.

Table 3. Measured CEI scoring pipeline costs on synthetic Barabási–Albert dependency graphs at five scales (60 = production-sized; 10^3 – 10^5 = synthetic). Centrality recompute occurs only on topology change; per-step costs occur on every scoring cycle. All measurements on a single CPU core; sampled betweenness uses $k = \max(20, \lfloor 5 \log_2 N \rfloor)$ pivots.

N	$ E $	$\bar{d}$	Betw. (s)	PageRank (s)	Per-step (ms)	Adj. mem (MB)
60	224	7.5	0.003	0.002	0.17	0.01
1,000	5,964	11.9	0.097	0.007	0.64	0.19
10,000	89,919	18.0	3.22	0.14	4.96	2.82
50,000	499,900	20.0	40.04	0.89	25.16	15.64
100,000	1,099,879	22.0	121.68	1.96	51.16	34.33

Three observations from Table 3. First, centrality recompute time scales roughly as $O(N \cdot k \cdot \log N)$, consistent with the theoretical complexity of k -sample betweenness on sparse graphs; the empirical scaling from $N = 10^4$ to $N = 10^5$ is approximately $38\times$ for a $10\times$ increase in nodes, between the linear and quadratic regimes. Second, the per-step cost (entropy plus CEI score plus governance lookup) scales sub-quadratically and remains below 100 ms even at the largest tested scale, leaving 99%+ of the 60-second telemetry budget available for I/O and orchestration. Third, sparse-adjacency memory grows linearly at approximately 0.35 KB per node, putting a 10^6 -node deployment at ~ 350 MB of adjacency memory—well within commodity server budgets. Federated regional decomposition becomes attractive above $N \approx 10^5$ where centrality recompute exceeds the recalibration interval; below that scale, single-node operation is comfortable.

Per-Component Runtime Overhead Breakdown. Table 4 decomposes the per-cycle latency at $N = 10^5$ into its constituent operations, showing where the budget is spent and where optimization effort would have leverage. Centrality recompute dominates the cold path (123.6 s, but amortized), while the warm path (per-step scoring) is dominated by the CEI scalarization and governance enforcement (37 ms) rather than entropy (14 ms).

Table 4. Per-component runtime overhead breakdown at $N = 10^5$ nodes, $|E| \approx 1.1 \times 10^6$ edges, on a single CPU core. “Cold” path runs only on topology change; “warm” path runs every scoring cycle.

Path	Component	Wall-clock	Frequency
Cold	Graph build (CSR sparse)	2.7 s	topology change
Cold	Sampled betweenness ($k = 83$)	121.7 s	topology change
Cold	PageRank ($\alpha = 0.85$)	2.0 s	topology change
Warm	Sliding-window entropy estimation	14.2 ms	per cycle
Warm	CEI scalarization + alloc projection	36.9 ms	per cycle
Warm	Governance floor enforcement	< 1 ms	per cycle
Total	warm path per cycle	51.2 ms	per cycle

The breakdown supports two operational claims. First, the warm path consumes approximately 0.085% of the 60-second telemetry interval at 10^5 -node scale, which means the framework is bandwidth-bound rather than compute-bound under realistic deployment conditions. Second, the cold path (centrality recompute) is dominated by sampled betweenness; replacement with a lower-cost centrality proxy (e.g., approximate k -core decomposition or local clustering coefficient) would reduce cold-path latency by an order of magnitude if topology change frequency required it. In practice, dependency topology in enterprise environments changes on minutes-to-hours timescales rather than per-cycle, so amortized cold-path cost is small relative to the warm-path budget.

Operational Validation Boundary. The measurements above were collected on synthetic graphs and a single CPU core. They do not constitute empirical validation at production scale; they establish that the scoring pipeline’s wall-clock cost remains tractable at 10^5 -node scales on commodity hardware. Empirical evaluation on production Kubernetes federation deployments spanning multiple clusters is the natural extension of this work.

11 Limitations and Future Work

This evaluation was conducted in a single production environment, limiting generalizability claims. The dependency graph construction relied on service mesh telemetry and network flow analysis, which may underrepresent implicit dependencies in environments with limited observability instrumentation. The governance policy store was populated manually from existing documentation, a process that may introduce classification errors in environments with incomplete or outdated governance documentation.

Adversarial Telemetry and Trust Assumptions. The CEI framework as evaluated here assumes trustworthy telemetry streams and accurate dependency-graph construction. In adversarial or zero-trust deployment contexts, several attack

surfaces require explicit mitigation. *Telemetry poisoning*—where a compromised node injects fabricated utilization metrics—can distort the entropy component and induce premature scale-up or consolidation decisions. *Dependency spoofing*—where attackers falsify service mesh records to inflate the centrality score of low-value nodes or deflate the score of high-value targets—can distort the centrality component. *Incomplete observability* from compromised collectors creates blind spots where the framework cannot detect emerging volatility. Robust extensions of the framework should consider Byzantine-tolerant telemetry aggregation [29], signed dependency-graph attestations, and outlier-resilient entropy estimators [30]. These mitigations are not currently implemented and represent the most operationally significant open question for hardened CEI deployments in defense or critical-infrastructure contexts.

The calibrated counterfactual analysis in Section 7 uses a synthetic environment parameterized from the deployment specifications, not a replay of production telemetry. The agreement on oscillation reduction (38.0% simulated vs 35–40% empirical) is strong, but the cost-reduction gap (22.7% simulated vs 27% empirical) indicates the synthetic environment does not capture all production-specific optimization opportunities. Trace-replay evaluation against the actual 90-day Azure telemetry log would close this gap and represents the highest-priority empirical extension.

The Lagrangian-PPO baseline in Section 7 did not converge to governance compliance within the 60,000-timestep training budget. The continued growth of the dual variable indicates the optimization is not at a saddle point; longer training or Constrained Policy Optimization [12] would likely achieve compliance. The reported failure of Lagrangian-PPO to converge within a practical training budget is itself an operationally meaningful finding—constrained-RL alternatives to CEI carry training-time costs that analytical allocation does not—but the convergent behavior at larger budgets remains to be characterized.

The adaptive weight recalibration mechanism demonstrated convergence limitations in four oscillation suppression cases where sustained demand variability exceeded the model’s adaptation speed. Formal convergence analysis of the recalibration update rules under adversarial demand conditions remains as future work. Multi-site validation across heterogeneous cloud environments and organizational contexts is the primary planned extension of this evaluation.

Integration with container orchestration platforms at scale, including Kubernetes-native operator patterns and service mesh policy enforcement, represents a practical deployment extension that would enable the framework to operate closer to the control plane rather than as an external recommendation engine. This integration pattern would reduce latency between CEI scoring and modification execution, potentially improving oscillation suppression responsiveness.

The LSTM workload forecaster in Section 8 was evaluated on synthetic workload data calibrated to deployment characteristics. Training the same model on the actual production telemetry trace would provide empirical forecasting accuracy values and would allow direct evaluation of the predictive-CEI integration

on real workloads. Other prediction architectures (Transformer, temporal convolution, state-space models) are natural alternatives to evaluate.

12 Conclusion

This paper presented the federated deployment architecture and 90-day empirical evaluation of a Centrality-Entropy Index framework for governance-aware cloud resource allocation in a production enterprise environment. The empirical evaluation produced quantified evidence of 27% cost reduction, 35–40% oscillation reduction, and 18–22% utilization efficiency improvement, with governance constraint filters activating on 23% of recommended modifications to preserve operational integrity. Comparative analysis confirmed that governance-unaware vendor tooling would have recommended operationally inappropriate modifications in approximately one third of cases.

The empirical results are supplemented by two methodological contributions: a calibrated counterfactual analysis comparing CEI against four alternative allocators including unconstrained PPO and Lagrangian-PPO, and an LSTM-based workload forecaster demonstrating predictive integration with the CEI entropy component. The counterfactual analysis confirms that unconstrained reinforcement learning collapses governance compliance to 0% in the cloud domain, replicating a soft-objective failure mode previously documented in NC3 and sensor-network domains [13,14]—a cross-domain consistency that argues for governance as a hard constraint rather than a reward term in any cyber-physical allocation system. The LSTM forecaster achieves a 60.7% mean skill score over naive persistence and identifies 15.2% of nodes per time-step where predictive entropy exceeds historical entropy, offering a clear integration target for predictive CEI extensions.

The framework’s platform-agnostic design and configurable governance layer support transferability to defense and public-sector environments facing structurally analogous governance challenges at larger scale. A provisional patent application has been filed describing aspects of the system architecture.

Declaration on Generative AI

The author used generative AI tools (Anthropic Claude) to assist with drafting and language refinement, manuscript revision, code scaffolding for simulation harnesses and the LSTM workload forecaster, and figure preparation. All scientific ideas, experimental design, deployment architecture decisions, results interpretation, and conclusions are the work of the human author. The author critically reviewed and edited all AI-assisted content and takes full responsibility for the manuscript’s accuracy, originality, and integrity.

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